Zhanna Kaufman

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Northampton, Massachusetts, USA

Objective

Seeking a research position. I have experience in formal verification, machine learning (including explainability), data analysis, and human study design. I also have experience as a research and innovation engineer in industry. I have passed my dissertation proposal and am aiming to graduate by Spring 2027.

Education

- **University of Massachusetts Amherst** *September 2021 - Current*
  PhD Candidate for Computer Science at Manning College of Information and Computer Sciences (CICS) Amherst, MA
  - Working under professor Yuriy Brun
  - Member of Laboratory for Advanced Software Engineering Research (LASER)
  - Research Assistant since June 2022
- **Worcester Polytechnic Institute** *July 2018*
  Masters of Science in Computer Science Worcester, MA
  - GPA: 3.91
- **Boston University** *June 2015*
  Bachelor of Science in Electrical Engineering with Minor in Computer Engineering Boston, MA
  - GPA: 3.65 (Magna Cum Laude)

Peer Reviewed Publications

C=Conference, J=Journal, *=Co-First Author

[C.1] **Zhanna Kaufman**, Emily First, Alex Sanchez-Stern, Kyle Thompson, Sorin Lerner, Yuriy Brun (2026). **ProofCoop: Collaborative Automated Formal Verification**. *ICSE 2026*. DOI: 10.1145/3744916.3787783

[C.2] **Zhanna Kaufman**, Madeline Endres, Cindy Xiong Bearfield, Yuriy Brun (2025). **Your Model Is Unfair, Are You Even Aware? Inverse Relationship Between Comprehension and Trust in Explainability Visualizations of Biased ML Models**. In *IEEE Vis 2025*. DOI: 10.1109/TVCG.2025.3634245

[C.3] Alex Sanchez-Stern, Abhishek Varghese, **Zhanna Kaufman**, Dylan Zhang, Talia Ringer, Yuriy Brun (2025). **QEDCartographer: Automating Formal Verification Using Reward-Free Reinforcement Learning**. In *IEEE/ACM 47th International Conference on Software Engineering (ICSE)*. DOI: 10.1109/ICSE55347.2025.00033

[C.4] Aimen Gaba*, **Zhanna Kaufman***, Jason Cheung, Marie Shvakel, Kyle Wm. Hall, Yuriy Brun, Cindy Xiong Bearfield (2023). **My Model is Unfair, Do People Even Care? Visual Design Affects Trust and Perceived Bias in Machine Learning**. In *IEEE VIS 2023*. DOI: 110.1109/TVCG.2023.3327192

[C.5] Arpan Agrawal, Emily First, **Zhanna Kaufman**, Tom Reichel, Shizhuo Zhang, Timothy Zhou, Alex Sanchez-Stern, Talia Ringer, Yuriy Brun (2023). **PRoofster: Automated Formal Verification**. In *2023 IEEE/ACM 45th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)*. DOI: 10.1109/ICSE-Companion58688.2023.00018

[J.1] Alex Sanchez-Stern, Emily First, Timothy Zhou, **Zhanna Kaufman**, Yuriy Brun, Talia Ringer (2023). **Passport: Improving Automated Formal Verification Using Identifiers**. In *IACM Transactions on Programming Languages and Systems*. DOI: 10.1145/3593374

Submitted

C=Conference, J=Journal, *=Co-First Author

[C.6] **Zhanna Kaufman**, Yuriy Brun, Adithya Murali, Madeline Endres (2026). **Programmers Are Poor and Overconfident Judges of LLM-Generated Assertions**.

Current Projects

- **Investigating developer understanding of LLM-generated formalizations of developer intent** *May 2025 - Current*
  Tools: ChatGPT, Python, Qualtrics, R
  - Completed a study on developer understanding of LLM-generated postconditions.
  - Identified a gap where developers struggle to indentify incorrect postconditions, and LLM-generated comments do not help.
  - Currently working on next steps to develop an intervention that facilitates such identification.
- **Effectively Measuring Trust in Chatbots** *September 2024 - Current*
  Tools: Qualtrics, Prolific, PyTorch, ChatGPT API, RPG Maker
  - Created a customizable API-based tool that allows developers to measure appropriate reliance on software systems with chatbot components using established procedures from HCI, Psychology, and Economics.
  - Gamified the tool to include multiple customizable settings and setting-specific scenarios to promote engagement and realistic decision-making.
  - Currently working on publication.
- **Effect of AI-generated facial features on trust** *May 2025 - Current*
  Tools: Google Colab, PyTorch, encoder4editing, Empirica, thispersondoesnotexist
  - Created a dataset from 25 female and 25 male faces obtained using thispersondoesnotexist.
  - Augmented the dataset using encoder4editing for a final set of 50 male presenting faces, 50 female presenting faces, and 50 androgynous faces.
  - Used Empirica to create a website from where our team can run trust games to test differences in trust toward faces with different feature presentations.
  - Currently drafing a publication.

Industry Experience

- **Google [🌐]** *September 2026 - November 2026*
  Research Scientist Intern San Francisco, CA
  - Investigated drivers of success and failure in human-AI collaborative software development trajectories.
  - Compared how these drivers differ and overlap between feature development and maintenance tasks.
  - Developed a maintenance overhead metric to quantify existing maintenance work and identify opportunities for agentic automation.

- **dModel [🌐]** *May 2026 - August 2026*
  Summer Research Intern San Francisco, CA
  - Built reinforcement learning environments for training frontier language models.
  - Focused on environments targeting interpretability and AI safety capabilities.
  - Performed weekly literature reviews to identify capability gaps and translate them into new environments.

- **The MITRE Corporation [🌐]** *June 2015 - January 2022*
  Senior Cyber Research and Innovation Engineer Burlington, MA
  - Developed vulnerability models and automated detection and remediation methods for network prototypes; investigated Windows API hooking and DLL injection to subvert spyware, and verified CVE mitigations for container file systems.
  - Designed solutions for messaging and communication across diverse architectures, network fabrics, and devices in a restricted environment; authored a white paper on internet protocol applications for a sponsor problem.
  - Built a DevSecOps pipeline automating builds, integration testing, and integrity checks for critical legacy software; integrated new motion trajectory calculations into a high-visibility system with full backwards compatibility.

Skills

- **Languages Used Regularly:** Python, R, JavaScript, TypeScript, OCaml, Rust, Bash
- **Languages Used in the Past:** Java, C/C++, Verilog
- **Data Science & Machine Learning:** PyTorch, sklearn, SciPy, TensorFlow, huggingface
- **Cloud Technologies:** Colab, Jupyter
- **DevOps & Version Control:** Git, Docker
- **Mathematical & Statistical Tools:** Rocq, Rstudio, Matlab, NumPy, Pandas, Matplotlib
- **Other Tools & Technologies:** HTML/CSS, Qualtrics, Tableau, LaTeX, Empirica
- **Research Skills:** Formal Verification, Machine Learning Explainability, Quantitative Analysis, Qualitative Analysis, IRB, Data Science, Prototyping, Human Study Design

Recent Honors and Awards

- **Dissertation and Thesis Proposal Writing Fellowship** *Spring 2025*
  Manning College of Information and Computer Sciences (CICS) [🌐]
  ◦ One of six students selected
- **SPECS Summer School** *Summer 2023*
  Tel Aviv University [🌐]
  ◦ Summer School on Formal Specifications and Analysis for Software Engineers

Mentorship and Teaching Experience

- **Long-term mentor for Undergraduate student** *August 2024 - Current*
  UMass Amherst LASER Lab [🌐]
  ◦ Co-lead a project and paper (currently under peer review) with junior (now senior) student in LASER
  ◦ Guided the student in learning the process of conducting research and writing papers
  ◦ Bridged technical knowledge gaps in software development and statistical analysis
- **Mentor for Undergraduate Research Volunteers (URV) program** *June 2024 - August 2024*
  UMass Amherst CICS Department [🌐]
  ◦ Volunteered as a PhD mentor for three undergaduate students performing research
  ◦ Developed a research project for each student that matched their research interests
  ◦ Students presented project results at an end of summer poster session
- **Teaching Assistant** *September 2021 - June 2022*
  UMass Amherst CICS Department
  ◦ TA'd for Computer Networks (CS453) and Introduction to Algorithms (CS311)

Volunteer Experience

- **Union Steward and Steward CoChair for CICS Department** *2022 - 2023*
  Graduate Employee Organization [🌐]
  ◦ Provided resources to CICS students to support their rights as Massachusetts workers
  ◦ Assisted in growing the union by encouraging membership through events and office hours
  ◦ As Steward CoChair, led regular meetings for department stewards to voice student needs
- **Red Cross Disaster Action Team member** *2020 - 2021*
  Red Cross [🌐]
  ◦ Provided families impacted by house or apartment fires with resources and financial aid
  ◦ Created a calm and compassionate environment where victims could voice their needs
  ◦ Connected victims with further long-term Red Cross support
- **live blue™ Service Corps Volunteer** *2016 - 2020*
  New England Aquarium [🌐]
  ◦ Assisted in environmental field projects throughout the greater Boston area
  ◦ Worked with the Parks Department to clean up local protected areas and remove invasive species
  ◦ Participated in educational and holiday events at the New England Aquarium

Additional Information

- **Languages:** Russian (Native speaker)
- **Hobbies:** Puzzle games, Slow biking, Even slower hiking, Horror games and movies